MTH321

Numerical Analysis of ODEs and PDEs

Financial SDEs

GBM benchmark and non-affine volatility

Project 1 Presentation

Anatomy of a stochastic differential equation

$$\mathrm{d}X_t = \underbrace{a(t, X_t)\,\mathrm{d}t}_{\text{drift}} + \underbrace{b(t, X_t)\,\mathrm{d}W_t}_{\text{diffusion}}$$

$$\Delta W_n = \sqrt{\Delta t}\, Z_n, \qquad Z_n \sim N(0,1)$$

GBM: $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t.$

Local interpretation

$$\mathbb{E}[\,\mathrm{d}X_t \mid \mathcal{F}_t] = a(t, X_t)\,\mathrm{d}t,$$
$$\mathrm{Var}(\,\mathrm{d}X_t \mid \mathcal{F}_t) = b(t, X_t)^2\,\mathrm{d}t.$$

- Drift controls the conditional mean movement.
- Diffusion controls the size of short-time random movement.

Itô's formula and the exact GBM solution

Itô's formula: if $\mathrm{d}S_t = a(t,S_t)\,\mathrm{d}t + b(t,S_t)\,\mathrm{d}W_t$ and $f \in C^{1,2}$, then

$$\boxed{\mathrm{d}f(t,S_t) = \left(f_t + af_S + \tfrac{1}{2}b^2 f_{SS}\right)\,\mathrm{d}t + bf_S\,\mathrm{d}W_t.}$$

Apply it to GBM

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t.$$

For $f(S) = \log S$, $f_t = 0$, $f_S = 1/S$, $f_{SS} = -1/S^2$:

$$\begin{aligned}\mathrm{d}\log S_t &= \left(\mu S_t\frac{1}{S_t} - \frac{1}{2}\sigma^2 S_t^2\frac{1}{S_t^2}\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t \\ &= \left(\mu - \tfrac{1}{2}\sigma^2\right)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.\end{aligned}$$

Integrate and exponentiate

$$\log S_t = \log S_0 + (\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t.$$

$$\boxed{S_t = S_0 e^{(\mu-\sigma^2/2)t+\sigma W_t}.}$$

$$\mathbb{E}[S_T] = S_0 e^{\mu T},$$

$$\mathrm{Var}(S_T) = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1).$$

The terminal law is lognormal; $S_t > 0$ for $S_0 > 0$.

Itô–Taylor expansion: which terms are retained?

For $\mathrm{d}U_t = a(U_t)\,\mathrm{d}t + b(U_t)\,\mathrm{d}W_t$, write $a_n = a(U_n)$, $b_n = b(U_n)$, and $b'_n = b'(U_n)$.

Scalar Itô–Taylor expansion (terms used by the two methods):

$$U_{t_{n+1}} = U_n + a_n h + b_n \Delta W_n + \tfrac{1}{2} b_n b'_n \big[(\Delta W_n)^2 - h\big] + R_n.$$

Euler–Maruyama: retain $a_n h$ and $b_n \Delta W_n$.

$$\boxed{U^{\mathrm{EM}}_{n+1} = U_n + a_n h + b_n \Delta W_n.}$$

Milstein: also retain the next stochastic correction.

$$\boxed{U^{\mathrm{Mil}}_{n+1} = U_n + a_n h + b_n \Delta W_n + \tfrac{1}{2} b_n b'_n \big[(\Delta W_n)^2 - h\big].}$$

R_n collects the omitted terms. The correction follows from $\int_{t_n}^{t_{n+1}} (W_s - W_{t_n})\,\mathrm{d}W_s = \tfrac{1}{2}[(\Delta W_n)^2 - h]$.

Log-price GBM: why EM and Milstein coincide

For $X_t = \log S_t$, Itô's formula gives

$$\mathrm{d}X_t = (\mu - \tfrac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Substitute the coefficients into the two methods:

$$a(X) = \mu - \tfrac{1}{2}\sigma^2, \qquad b(X) = \sigma \text{ (constant)}, \qquad \boxed{b'(X) = 0.}$$

$$X_{n+1}^{\mathrm{EM}} = X_n + (\mu - \tfrac{1}{2}\sigma^2)h + \sigma\Delta W_n,$$
$$X_{n+1}^{\mathrm{Mil}} = X_n + (\mu - \tfrac{1}{2}\sigma^2)h + \sigma\Delta W_n + \underbrace{\tfrac{1}{2}\sigma \cdot 0\big[(\Delta W_n)^2 - h\big]}_{=0}.$$

Both give the same exact grid update.

For the original price equation, $b(S) = \sigma S$ and $b'(S) = \sigma$: $bb' = \sigma^2 S \neq 0$ when $\sigma \neq 0$ and $S > 0$. We therefore compare EM and Milstein in price space.

Strong and weak convergence answer different questions

Strong error

$$\mathbb{E}\left|S_T^{(h)} - S_T\right| \leq Ch^p$$

Same Brownian path for the numerical and exact solutions.

$$p_{\mathrm{EM}} = \tfrac{1}{2}, \qquad p_{\mathrm{Mil}} = 1.$$

Weak error

$$\left|\mathbb{E}\phi(S_T^{(h)}) - \mathbb{E}\phi(S_T)\right| \leq Ch^q$$

For smooth observables, both methods have $q = 1$. For the GBM mean,

$$\mathbb{E}[S_N] = S_0(1+\mu h)^{T/h} = S_0 e^{\mu T} + O(h).$$

Strong convergence: calculation

1. **Fix T, choose $h = T/N$, and generate M independent paths:**

$$\Delta W_n^{(m)} = \sqrt{h}\, Z_n^{(m)}, \qquad Z_n^{(m)} \sim N(0,1).$$

2. **Compute numerical and exact prices on the same Brownian path:**

$$S_{n+1}^{\mathrm{EM}} = S_n^{\mathrm{EM}}(1 + \mu h + \sigma \Delta W_n),$$
$$S_{n+1}^{\mathrm{Mil}} = S_n^{\mathrm{Mil}}\left[1 + \mu h + \sigma \Delta W_n + \tfrac{1}{2}\sigma^2((\Delta W_n)^2 - h)\right],$$
$$S_T = S_0 \exp\left[(\mu - \tfrac{1}{2}\sigma^2)T + \sigma \sum_n \Delta W_n\right].$$

3. **Estimate terminal L^1 strong error (absolute value before averaging):**

$$e_s(h) = \mathbb{E}|S_{T,h} - S_T|, \qquad \widehat{e}_s(h) = \frac{1}{M}\sum_{m=1}^{M} |S_{T,h}^{(m)} - S_T^{(m)}|.$$

4. **Halve h repeatedly; couple grids by summing fine increments:**

$$e_s(h) \approx C_s h^{p_s} \implies \widehat{p}_s = \frac{\log[\widehat{e}_s(h)/\widehat{e}_s(h/2)]}{\log 2}.$$

Weak convergence: calculation

1. Choose an observable: here $\phi(s) = s$ (mean price). Use the numerical and exact endpoints generated on the preceding slide.

$$b(h) = \mathbb{E}[S_{T,h}] - \mathbb{E}[S_T], \quad \widehat{b}(h) = \frac{1}{M}\sum_m (S_{T,h}^{(m)} - S_T^{(m)}), \quad \widehat{e}_w(h) = |\widehat{b}(h)|.$$

2. Derive the exact GBM mean bias for both EM and Milstein:

$$\mathbb{E}[\Delta W_n] = 0, \qquad \mathbb{E}[(\Delta W_n)^2 - h] = 0 \implies \mathbb{E}[S_{n+1} \mid \mathcal{F}_{t_n}] = (1+\mu h)S_n.$$

$$\mathbb{E}[S_{T,h}] = S_0(1+\mu h)^{T/h}, \qquad b(h) = S_0(1+\mu h)^{T/h} - S_0 e^{\mu T}.$$

3. Expand for small h:

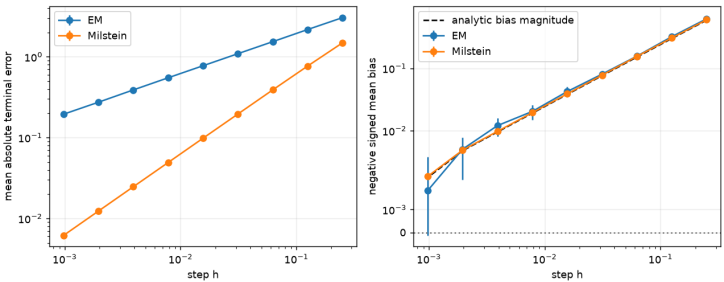
$$\frac{T}{h}\log(1+\mu h) = \mu T - \tfrac{1}{2}\mu^2 T h + O(h^2),$$

$$b(h) = -\tfrac{1}{2} S_0 e^{\mu T}\mu^2 T h + O(h^2).$$

Thus $e_w(h) = |b(h)| \sim C_w h$ for $\mu \neq 0$: **weak order one**.

4. Numerical check: repeat across grids and fit $\log \widehat{e}_w = \alpha + p_w \log h$ using resolved nonzero biases. **Weak convergence** concerns expectations of test functions; this derivation verifies the mean observable, not every payoff.

GBM convergence and Monte Carlo uncertainty



65,536 paths

Strong slopes:

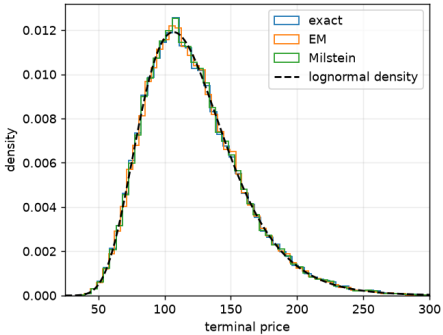
$$\widehat{p}_{\mathrm{EM}} = 0.4975,$$
$$\widehat{p}_{\mathrm{Mil}} = 0.9972.$$

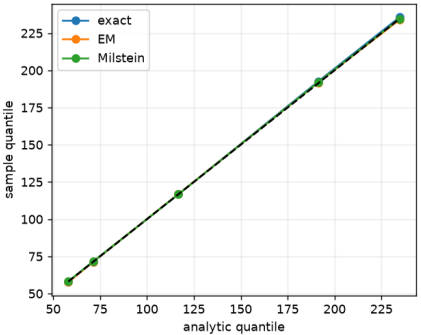
Weak-mean slopes:

0.9610 EM,
0.9948 Milstein.

Histogram overlay and quantile comparison

Benchmark: $S_0 = 100$, $\mu = 0.2$, $\sigma = 0.3$, $T = 1$; 65,536 paths, EM/Milstein with $N = 32$.





Histogram: simulated densities track the dashed exact lognormal curve, including its right-skewed shape.

Quantiles: x = exact, y = sampled. Points near $y = x$ show agreement; the next table reveals small tail differences.

Quantile comparison: centre and tails

Let $m_T = \log S_0 + (\mu - \sigma^2/2)T$ and $v_T = \sigma^2 T$. Then $\log S_T \sim N(m_T, v_T)$, giving the exact density and quantile:

$$f(s) = \frac{\exp[-(\log s - m_T)^2/(2v_T)]}{s\sqrt{2\pi v_T}}, \quad s > 0, \qquad q_p = \exp\big(m_T + \sqrt{v_T}\,\Phi^{-1}(p)\big)\,.$$

Simulation: use all 65,536 terminal samples. Histogram bin height is $count/(M \times \text{bin width})$; overlay $f(s)$ and compare sample quantiles.

p	Exact law	Exact simulation	EM ($N = 32$)	Milstein ($N = 32$)
0.01	58.106	58.032	57.644	58.280
0.05	71.287	71.348	71.312	71.565
0.50	116.766	116.740	116.877	116.724
0.95	191.259	192.508	191.771	191.907
0.99	234.645	235.721	233.999	234.685

Sample quantiles fluctuate even for exact simulation. These are descriptive checks, not simultaneous goodness-of-fit confidence statements.

Positivity: negative-update conditions

Assume $S_n > 0$, $h > 0$, $\sigma > 0$, and $\Delta W = \sqrt{h}Z$, $Z \sim N(0,1)$.

EM: a linear multiplier

$$S_{n+1} = S_n F_{\mathrm{EM}}, \quad F_{\mathrm{EM}} = 1 + \mu h + \sigma\sqrt{h}Z.$$

$$S_{n+1} < 0 \iff Z < -\frac{1+\mu h}{\sigma\sqrt{h}}.$$

$$p_{\mathrm{EM}}(h) = \Phi\left(-\frac{1+\mu h}{\sigma\sqrt{h}}\right) > 0.$$

An unbounded Gaussian lower tail means no unconditional positivity guarantee.

Milstein: complete the square

$$F_{\mathrm{Mil}} = 1 + \mu h + \sigma\sqrt{h}Z + \tfrac{1}{2}\sigma^2 h(Z^2 - 1)$$

$$= \tfrac{1}{2}\sigma^2 h\left(Z + \frac{1}{\sigma\sqrt{h}}\right)^2 + \underbrace{\tfrac{1}{2} + (\mu - \tfrac{1}{2}\sigma^2)h}_{c(h)}.$$

$c(h) \geq 0$: no negative multiplier.
$c(h) < 0$: negative multipliers occur between the two roots.

$$z_{\pm} = \frac{-1 \pm \sqrt{-2c(h)}}{\sigma\sqrt{h}}, \qquad p_{\mathrm{Mil}} = \Phi(z_+) - \Phi(z_-).$$

Milstein is positive almost surely for our baseline parameters, but fails for some high-volatility, coarse-step settings. Neither scheme is unconditionally positivity-preserving.

When are negative prices possible?

Assume $S_n > 0$, $h = \Delta t > 0$, $\sigma > 0$, and $Z \sim N(0,1)$.

Euler–Maruyama (EM)

A negative update occurs when

$$Z < -\frac{1+\mu h}{\sigma\sqrt{h}}.$$

To avoid a negative update

$$Z \geq -\frac{1+\mu h}{\sigma\sqrt{h}}.$$

This condition depends on the random draw. Gaussian draws have no lower bound, so no finite $h > 0$ guarantees nonnegative prices for price-space EM when $\sigma > 0$.

Milstein

Negative updates are possible when

$$c(h) = \tfrac{1}{2} + (\mu - \tfrac{1}{2}\sigma^2)h < 0.$$

They occur for draws between the two roots shown on the preceding slide.

To guarantee no negative updates

$$\boxed{1 + (2\mu - \sigma^2)h \geq 0.}$$

If $\sigma^2 > 2\mu$, choose

$$\boxed{h \leq \frac{1}{\sigma^2 - 2\mu}.}$$

If $\sigma^2 \leq 2\mu$, every $h > 0$ satisfies the condition.

Positive-price alternative: simulate log price and set $S_n = e^{X_n} > 0$.

Non-affine stochastic volatility

$$
\begin{aligned}
\mathrm{d}X_t &= \left(\mu - \frac{1}{2} g(Y_t)^2\right)\,\mathrm{d}t + g(Y_t)\,\mathrm{d}W_t^{(1)},\\
\mathrm{d}Y_t &= \kappa(\theta - Y_t)\,\mathrm{d}t + \xi\sqrt{1+Y_t^2}\,\mathrm{d}W_t^{(2)},\\
g(y) &= \sigma_{\min} + (\sigma_{\max} - \sigma_{\min})\ell(y),\\
\ell(y) &= \frac{1}{1+e^{-y}},\\
\mathrm{d}\langle W^{(1)}, W^{(2)}\rangle_t &= \rho\,\mathrm{d}t,\\
S_t &= e^{X_t}.
\end{aligned}
$$

$\mu = 0.05$	expected return rate
$\kappa = 2$	speed of reversion
$\theta = -0.2$	long-run factor level
$\xi = 0.6$	factor noise scale
$\rho = -0.7$	return-volatility relation
$\sigma_{\min} = 0.10$	lower volatility bound
$\sigma_{\max} = 0.50$	upper volatility bound

Why the scalar Milstein formula does not transfer

Write the correlated noises using independent Brownian motions B_1, B_2:

$$W^{(1)} = B_1, \qquad W^{(2)} = \rho B_1 + \sqrt{1-\rho^2}\, B_2.$$

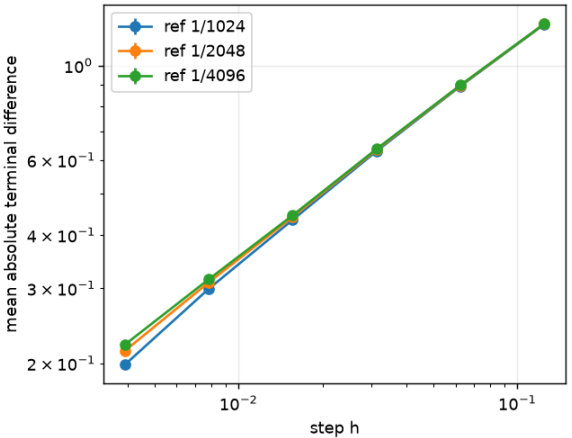
With $q(y) = \xi\sqrt{1+y^2}$, the diffusion vector fields are

$$b_1 = \begin{pmatrix} g \\ \rho q \end{pmatrix}, \qquad b_2 = \begin{pmatrix} 0 \\ \sqrt{1-\rho^2}\, q \end{pmatrix}.$$

Their commutator is

$$[b_1, b_2] = \begin{pmatrix} -\sqrt{1-\rho^2}\, q(y) g'(y) \\ 0 \end{pmatrix} \neq 0.$$

Nested Brownian grids validate the non-affine model

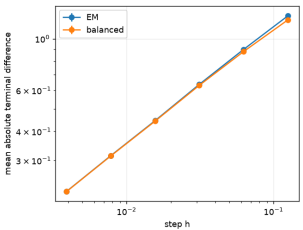


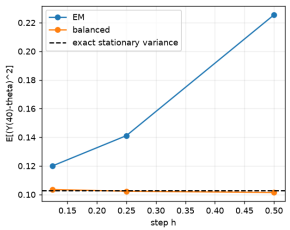
Coarse increments sum fine increments:

$$\Delta W_n^{(h)} = \sum_{j=0}^{r-1} \Delta W_{rn+j}^{(h_f)}.$$

- 32,768 paths
- References: 1024, 2048, 4096 steps
- Finest slope: 0.5003

Beyond the baseline: factor stability and variance reduction





For the call payoff $P = (S_T - K)^+$,

$$P_{\mathrm{cv}} = P - \widehat{\beta}\left(S_T - S_0 e^{\mu T}\right).$$

Independent pilot for $\widehat{\beta}$:

variance reduction $= 6.23\times$.

At matched GBM error:

$$N_{\mathrm{EM}} = 256, \qquad N_{\mathrm{Mil}} = 16.$$